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I. Project Objective

The Attendance Monitoring System is a mobile-based application designed to help educational institutions manage and monitor student attendance in a more efficient and organized manner. Developed using Flutter for the front-end and Firebase for the back-end, the system provides a digital alternative to traditional manual attendance methods. Flutter is commonly used for cross-platform mobile apps, while Firebase is widely used for authentication, cloud data storage, and real-time synchronization in attendance-style applications

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The application enables faculty members to mark attendance easily, allows students to view their attendance records, and provides administrators with the ability to manage users and oversee the overall system. Flutter-and-Firebase attendance app examples commonly use role-based access for students, faculty, and admins, along with centralized attendance data handling.

By digitizing the attendance process, the system improves accuracy, reduces paperwork, saves time, and offers a convenient way to access attendance information anytime through a mobile device. Firebase-backed attendance systems are commonly used because they support secure data handling and quick access to attendance records.

II. Purpose of the System.

The Attendance Monitoring System is developed to provide a digital and efficient way to manage student attendance in educational institutions using Flutter and Firebase. It replaces manual attendance methods with a mobile-based system that is faster, more accurate, and easier to manage. Digital attendance systems commonly use role-based access, centralized data handling, and real-time updates to improve attendance management.

Student

- View personal attendance records, subject-wise attendance, and attendance percentage.

Faculty

- Mark and manage student attendance for assigned subjects or classes.

Admin

- Manage users and monitor overall attendance data in the system.

System

- Store and sync attendance data in real time using Firebase.
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III. Modules

1. Authentication Module

- Handles student, faculty, and admin login using Firebase Authentication and role-based access through Firestore.

2. Dashboard Module

- Displays an overview of attendance details such as subject-wise attendance, total present and absent sessions, and attendance percentage.

3. Student Module

- Allows students to view their attendance records and subject-wise attendance status.

4. Faculty Module

- Allows faculty members to mark and manage attendance for assigned subjects or classes.

5. Subject Management Module

- Manages subject details, assigned faculty, and enrolled students.

6. Attendance Records Module

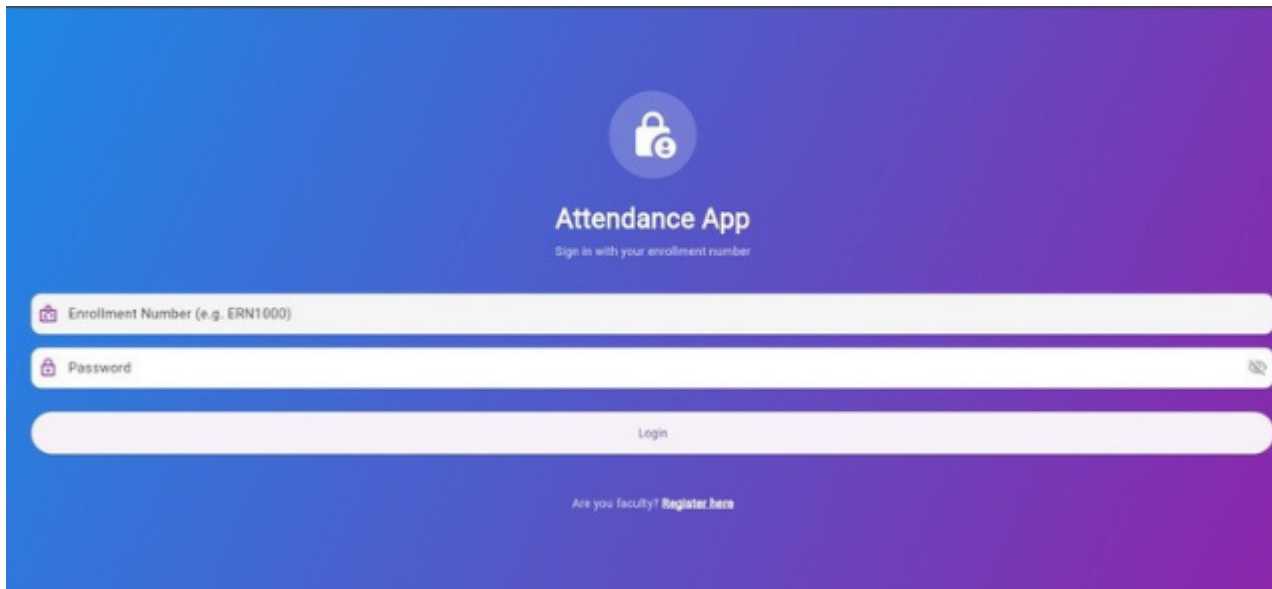
- Stores and displays attendance history with date, subject, and status.

7. Profile Module

- Allows users to view and update personal details and account information.

8. Admin Module

- Allows admin to manage students, faculty, subjects, and overall attendance data.

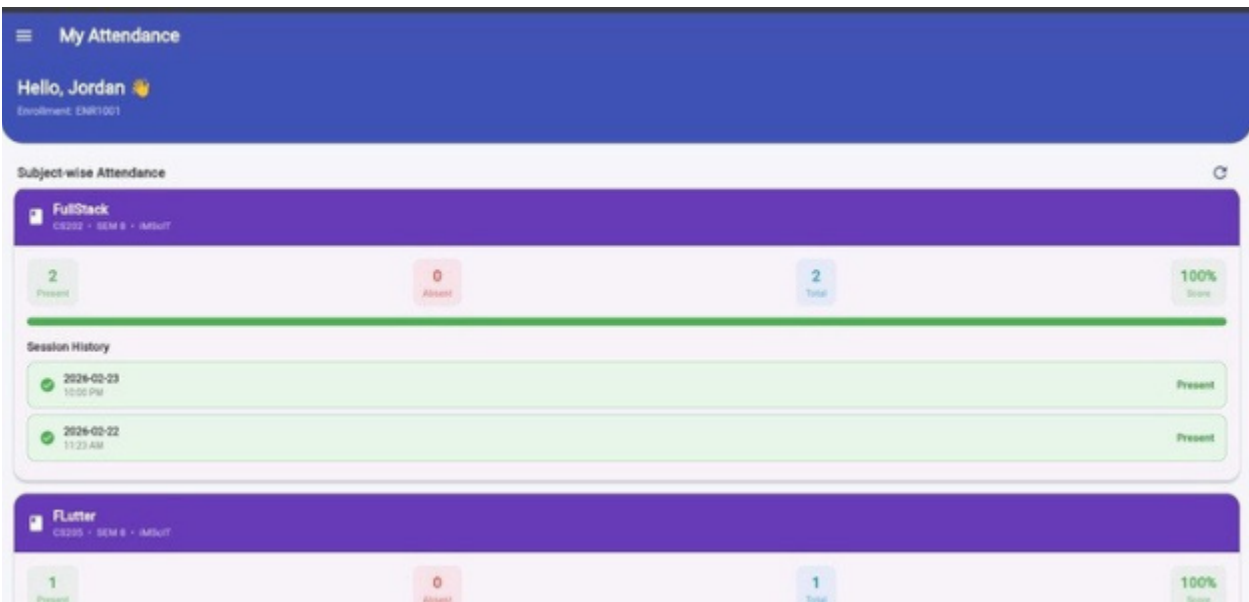


1. Student Dashboard – Attendance Overview

Description: This screen shows the Login Page where both students and faculty members can securely access the system. Students log in using their Enrollment Number, while faculty members log in using their Faculty ID.

Main Widgets Used:

Scaffold, Container, BoxDecoration, LinearGradient, Center, SingleChildScrollView, Column, TextField, IconButton, Icon, SizedBox, Text, ElevatedButton, Row, Expanded, GestureDetector



2. Student Dashboard – Attendance Overview

Description: This screenshows the student dashboard where the student can view subject-wise attendance. It displays the number of present and absent sessions along with the attendance percentage. The system also shows session history with date and attendance status.

Main Widgets Used:

Scaffold,AppBar,Drawer,Card, ListView, Text, LinearProgressIndicator, ContainerStreamBuilder,FutureBuilder, ClipRRect

← Edit Profile

J
ENR1001

Personal Details

Full Name
Jordan Blake

Email Address
black@gmail.com

Gender
☒ Male ☐ Female

Change Password (optional)

Current Password

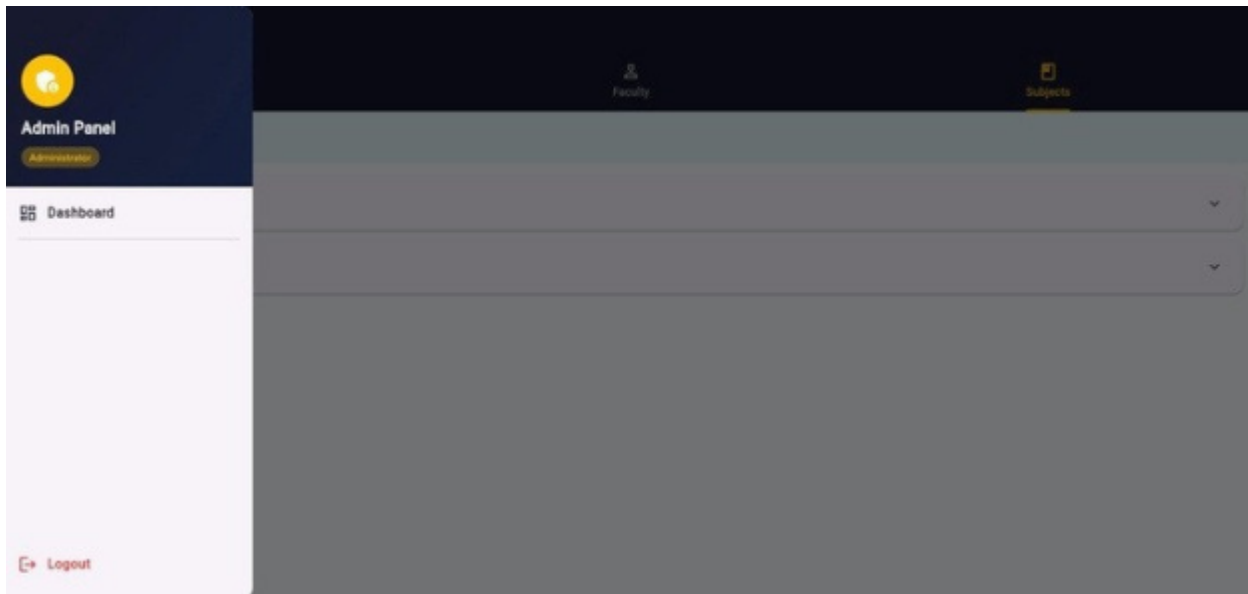
New Password

3. Student Profile / Edit Profile

Description: This screen allows students to view and update their profile information such as name, email, gender, and password. Students can modify personal details and save the updated information securely.

Main Widgets Used:

Scaffold, AppBar, TextFormField, RadioListTile, ElevatedButton, Container, SingleChildScrollView, Divider

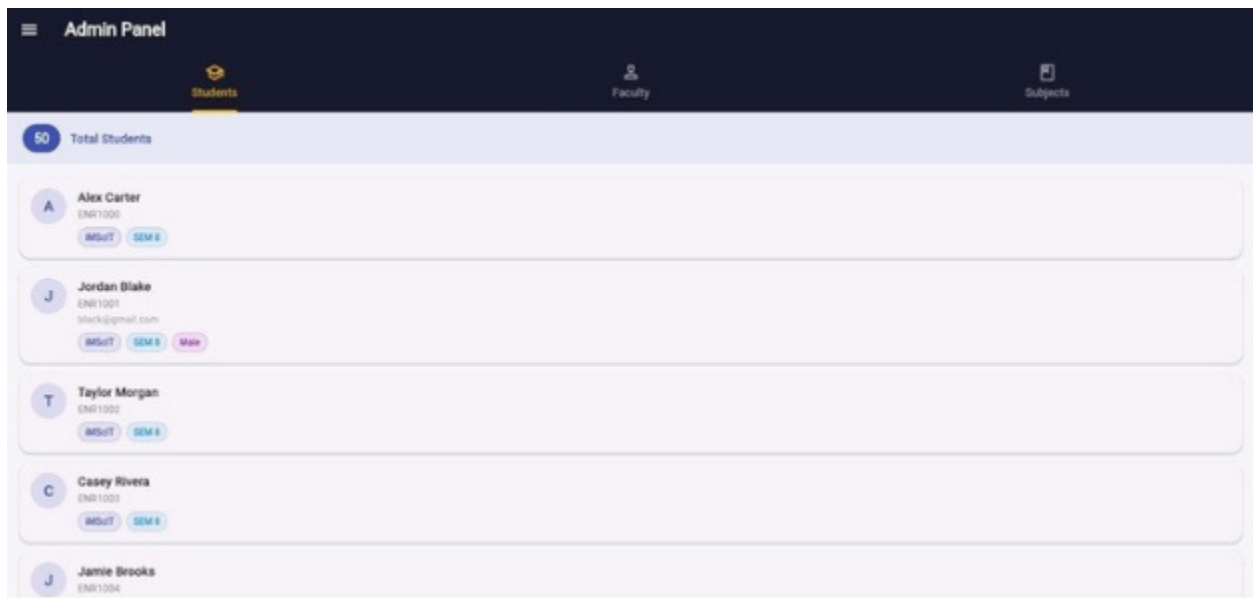


1. AdminDashboard

Description: The admin dashboard allows the administrator to manage the system. Admin can view information about students, faculty members, and subjects. It provides navigation options to manage different modules of the system.

Main Widgets Used:

Scaffold, Drawer, AppBar, GridView, Container, Icon, Text, StreamBuilder, CircleAvatar, Wrap, ListTile

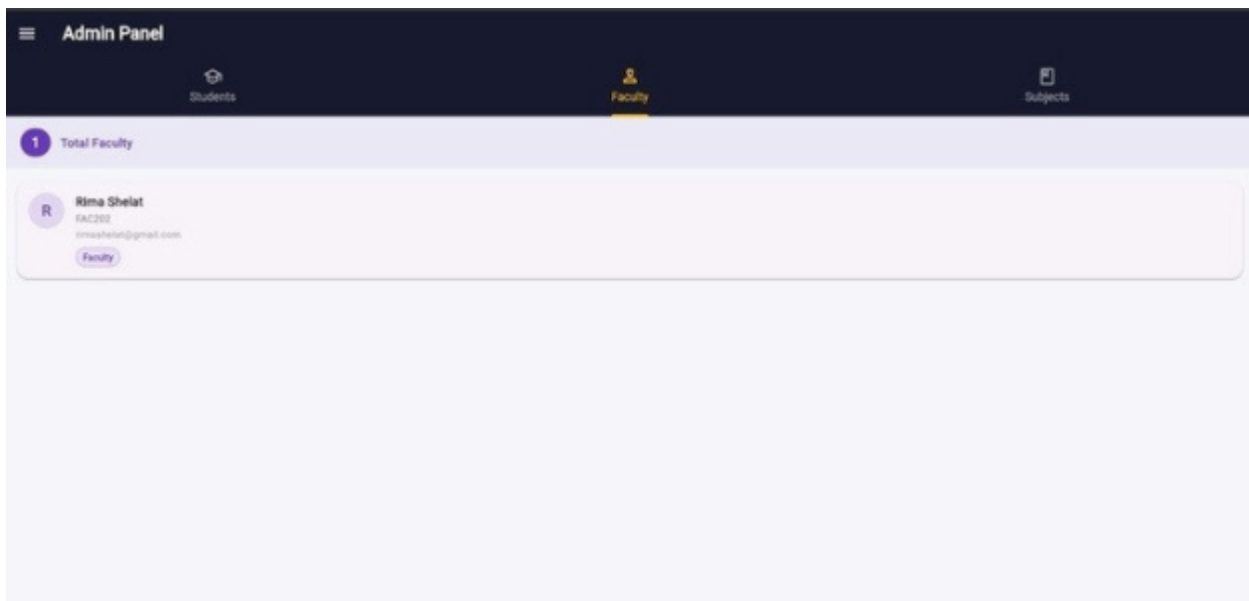


2 Manage Students

Description: This screen allows the admin to view and manage student records. The system displays the total number of students along with their enrollment number, name, course, and semester.

Main Widgets Used:

Scaffold, ListView, Card, Text, Icon, Container, StreamBuilder, CircleAvatar

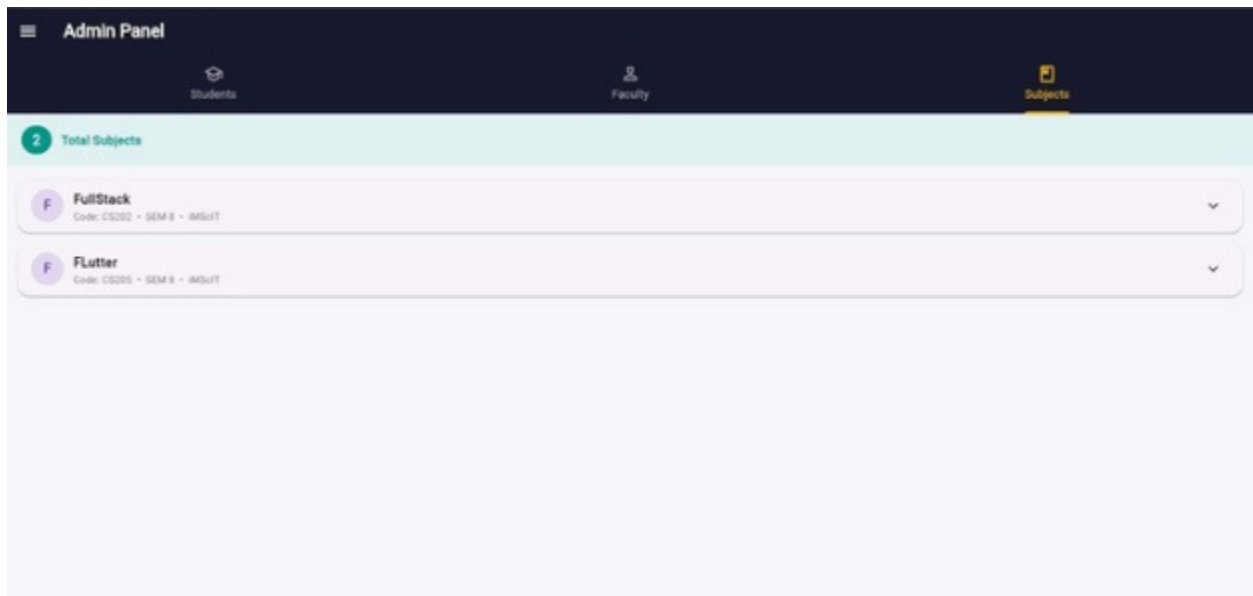


3. ManageFaculty

Description: This screen allows the admin to view faculty members registered in the system. Admin can check faculty details such as faculty ID, name, and email address.

Main Widgets Used:

Scaffold, ListView, Card, Text, Icon, Container, StreamBuilder

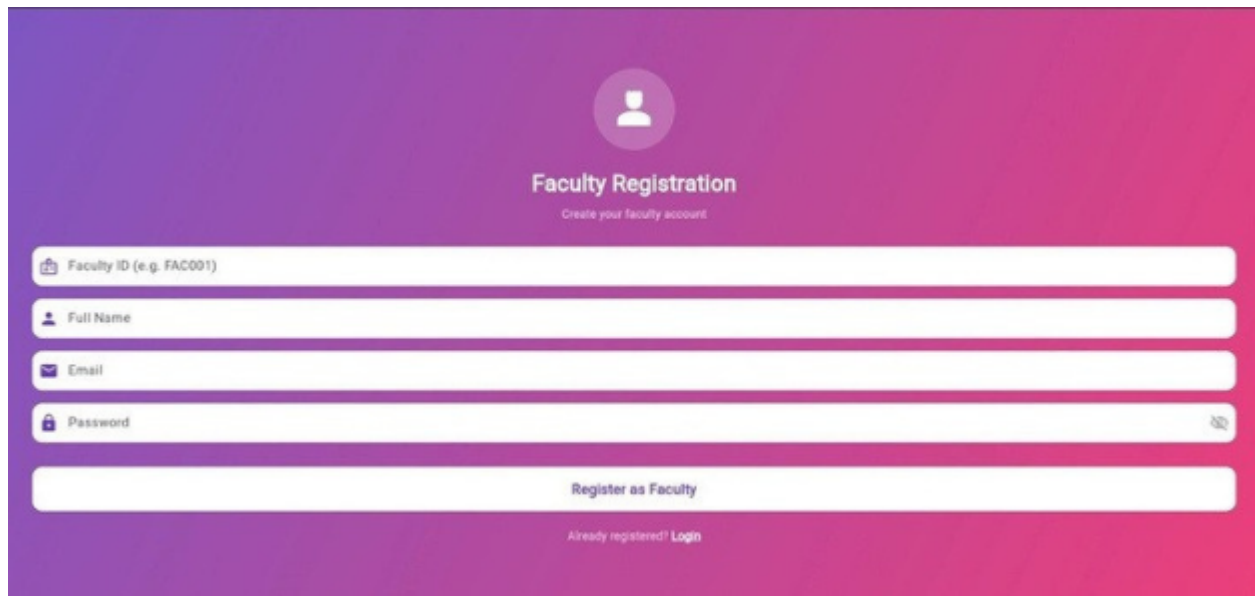


4. ManageSubjects

Description: This screen displays all subjects available in the system along with subject code, semester, and course information. Admin can monitor and manage subjects assigned to faculty members.

Main Widgets Used:

Scaffold, Card, ListView, Text, Container, ExpansionTile, StreamBuilder

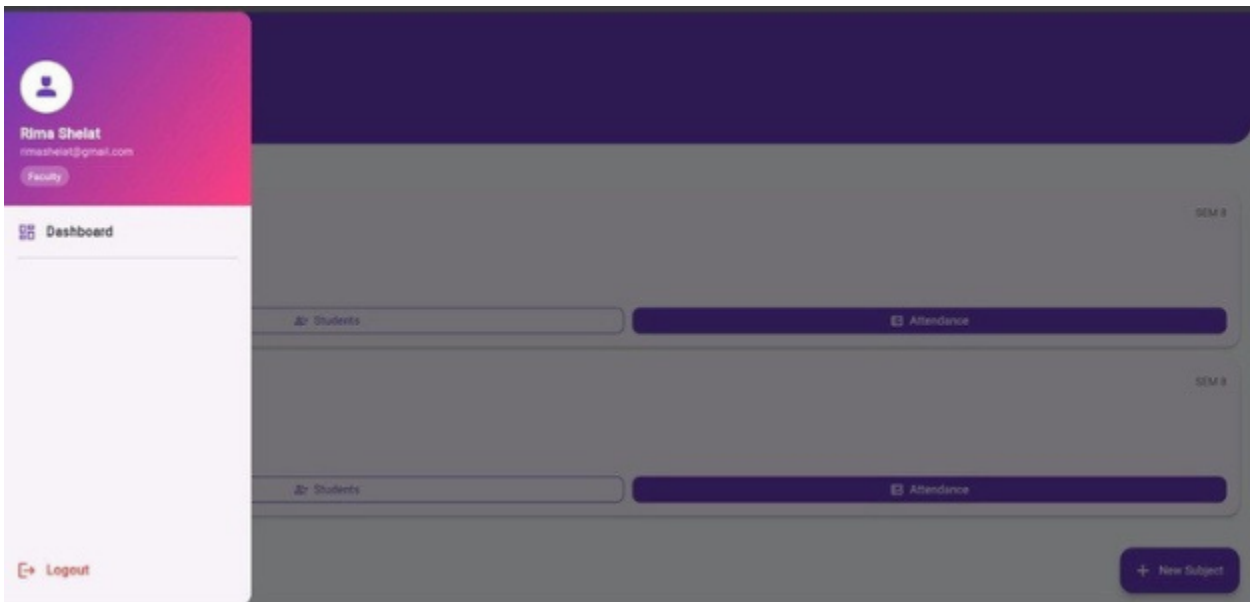
The image shows a mobile app interface for faculty registration. At the top, there is a purple-to-pink gradient background. In the center, there is a white circular icon containing a person silhouette. Below the icon, the text "Faculty Registration" is displayed in white, followed by the subtitle "Create your faculty account" in a smaller font. The form consists of four white input fields with icons on the left: a briefcase icon for "Faculty ID (e.g. FAC001)", a person icon for "Full Name", an envelope icon for "Email", and a lock icon for "Password". To the right of the password field is a small eye icon for toggling visibility. Below the input fields is a large white button with the text "Register as Faculty". At the bottom, there is a link that says "Already registered? Login".

Faculty Registration Page

Description: This screen allows new faculty members to create an account in the Attendance Monitoring System. The faculty member must enter details such as Faculty ID, Full Name, Email Address, and Password. After filling in the required information, the faculty can register by clicking the **Register as Faculty** button. Once registered, the faculty member can log in and access the faculty dashboard to manage subjects and mark student attendance.

Widgets Used:

Scaffold, Container, TextField, Icon, ElevatedButton, Column, Center, LinearGradient, SingleChildScrollView



1. Faculty Dashboard

Description: This screen shows the faculty dashboard where faculty members can view their assigned subjects and manage attendance. Faculty can navigate to student lists and attendance marking for each subject.

Main Widgets Used:

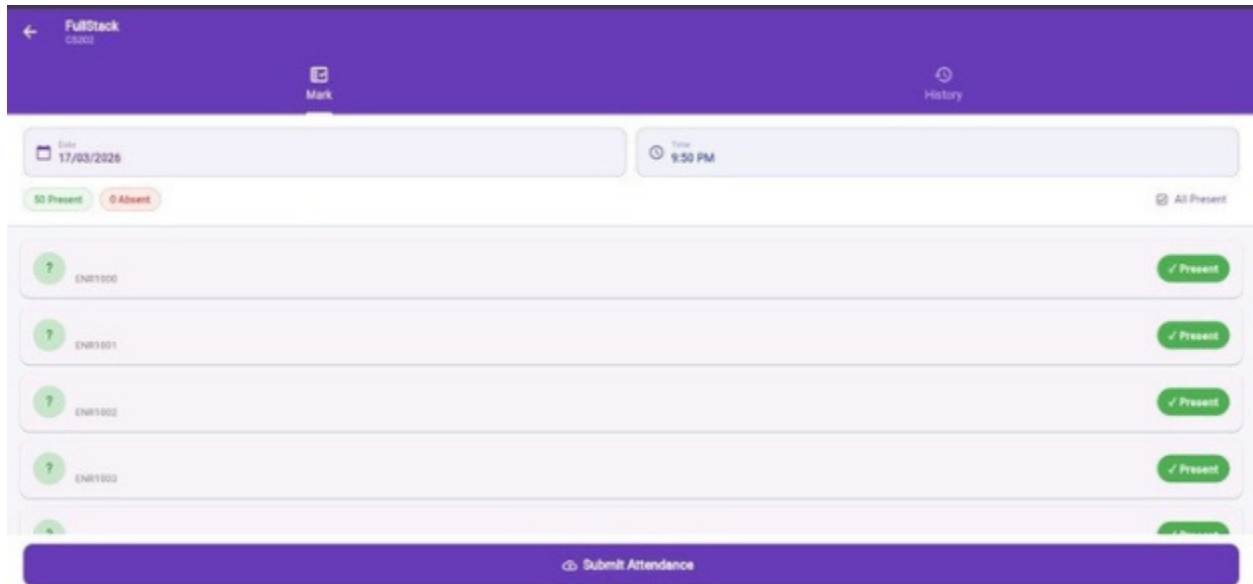
Scaffold, Card, ListView, ElevatedButton, Text, Drawer, FloatingActionButton.extended, StreamBuilder, FutureBuilder, CircleAvatar, ListTile

2. Create New Subject

Description: This screen allows faculty members to create a new subject by entering subject name, subject code, semester, and course details. Faculty can also upload student details using a CSV file.

Main Widgets Used:

TextFormField, DropdownButtonFormField, ElevatedButton, OutlinedButton (that triggers file_picker), Container, Scaffold, AppBar, SingleChildScrollView

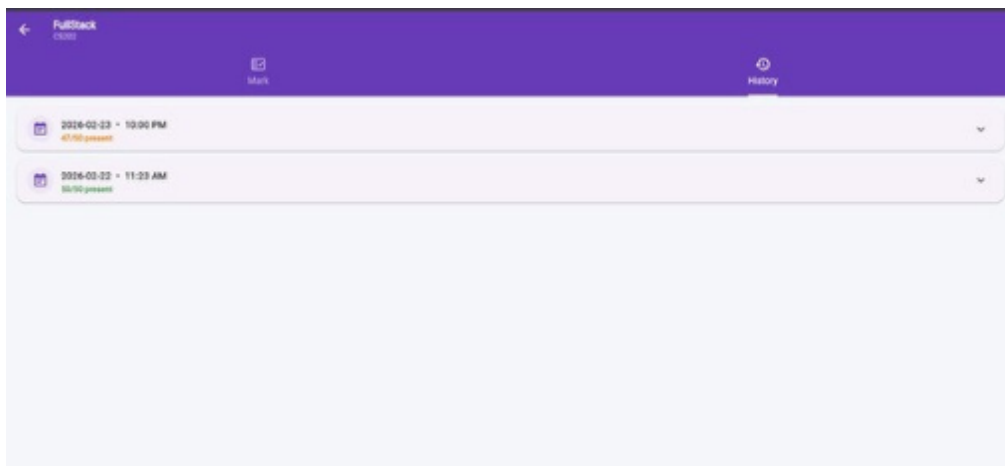


3. Mark Attendance

Description: This screen allows faculty members to mark attendance for students in a specific subject. Faculty can select present or absent status for each student and submit attendance for the session.

Main Widgets Used:

ListView, AnimatedContainer, Text, ElevatedButton, Container, Scaffold, AppBar, TabBar, TabBarView, DatePickerDialog, TimePickerDialog



4. Attendance History

Description: This screen allows facultymembers to seeattendance history for specific subject. Faculty can see total present student for the session including time and date .

Main Widgets Used: StreamBuilder, ListView.builder, Card, ExpansionTile, CircleAvatar, Icon, Text, Padding, Column, Row, Expanded, Divider, SizedBox, Center

2.4 Database

Faculty Collection:

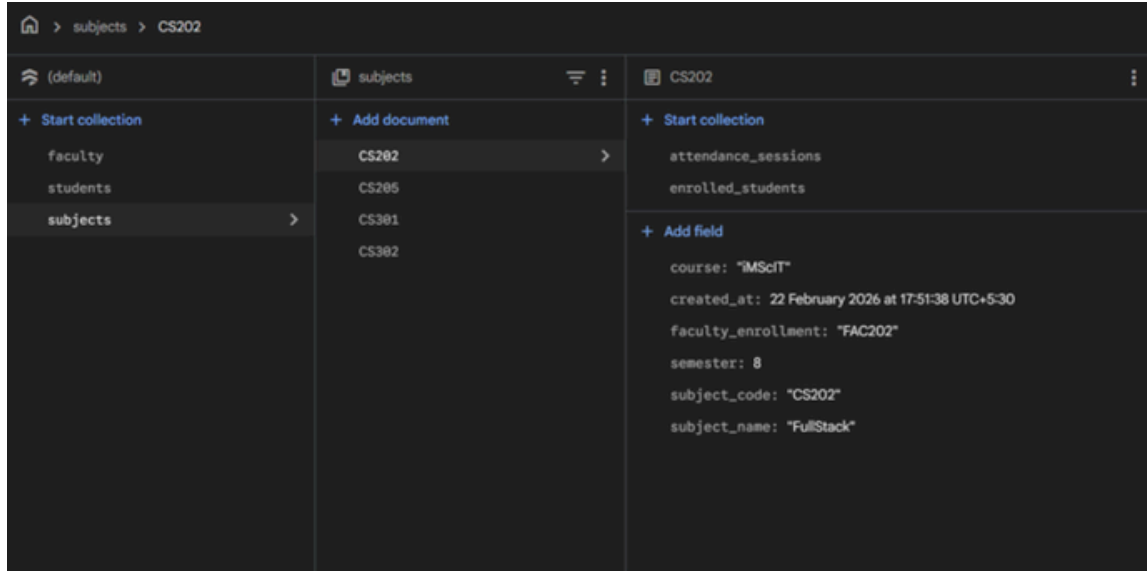
🏠 > faculty > FAC202 ✎		
🔄 (default)	📄 faculty	📄 FAC202
+ Start collection	+ Add document	+ Start collection
faculty >	FAC202 >	+ Add field
students	FAC21	created_at: 22 February 2026 at 17:49:58 UTC+5:30
subjects		email: "rimashehat@gmail.com"
		enrollment_no: "FAC202"
		name: "Rima Shelat"
		password: "123456"
		role: "Faculty"

Students Collection:

🏠 > students > ENR10001		
🔄 (default)	📄 students	📄 ENR10001
+ Start collection	+ Add document	+ Start collection
faculty	ENR10001 >	+ Add field
students >	ENR10002	course: "BCA"
subjects	ENR10003	created_at: 15 April 2026 at 12:44:28 UTC+5:30
	ENR10004	email: null
	ENR10005	enrollment_no: "ENR10001"
	ENR10006	gender: null
	ENR10007	name: "Aditi Nair"
	ENR10008	password: "pass123"
	ENR10009	role: "Student"
	ENR1001	semester: 5
	ENR10010	subject_ids
	ENR10011	0 "CS301"
	ENR10012	1 "CS303"

2.4 Database

Subjects Collection:



Attendance_sessions Collection:

